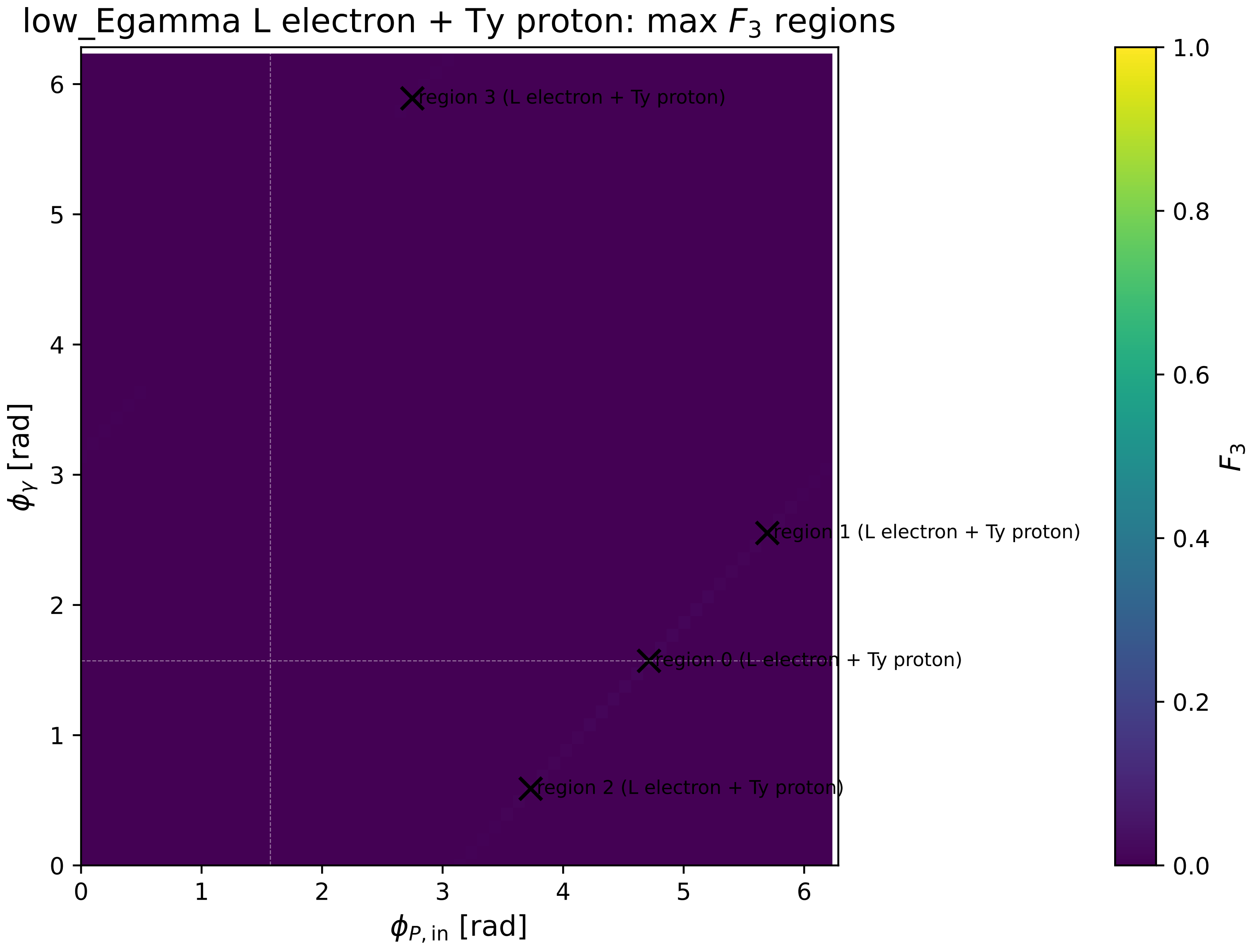
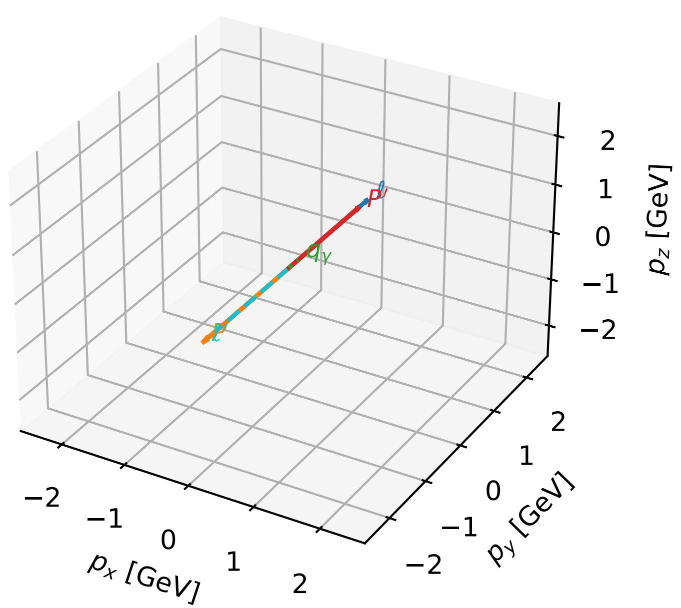


low_Egamma L electron + Ty proton: max F_3 regions
region 3 (L electron + Ty proton)
region 1 (L electron + Ty proton)
region 0 (L electron + Ty proton)
region 2 (L electron + Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
F_3
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

L electron + Ty proton characteristic max F_3 configuration

3D momenta

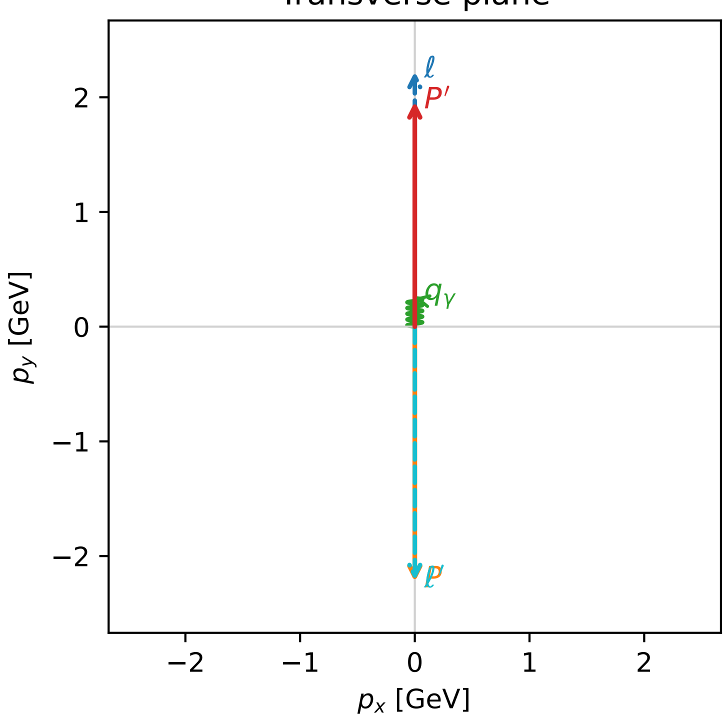



```
f3_low_egamma_lty_region_0  F_3=0.013528
region: low_Egamma_LTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
```
incoming state: $|L_e\rangle \otimes |T_{y,p}\rangle$

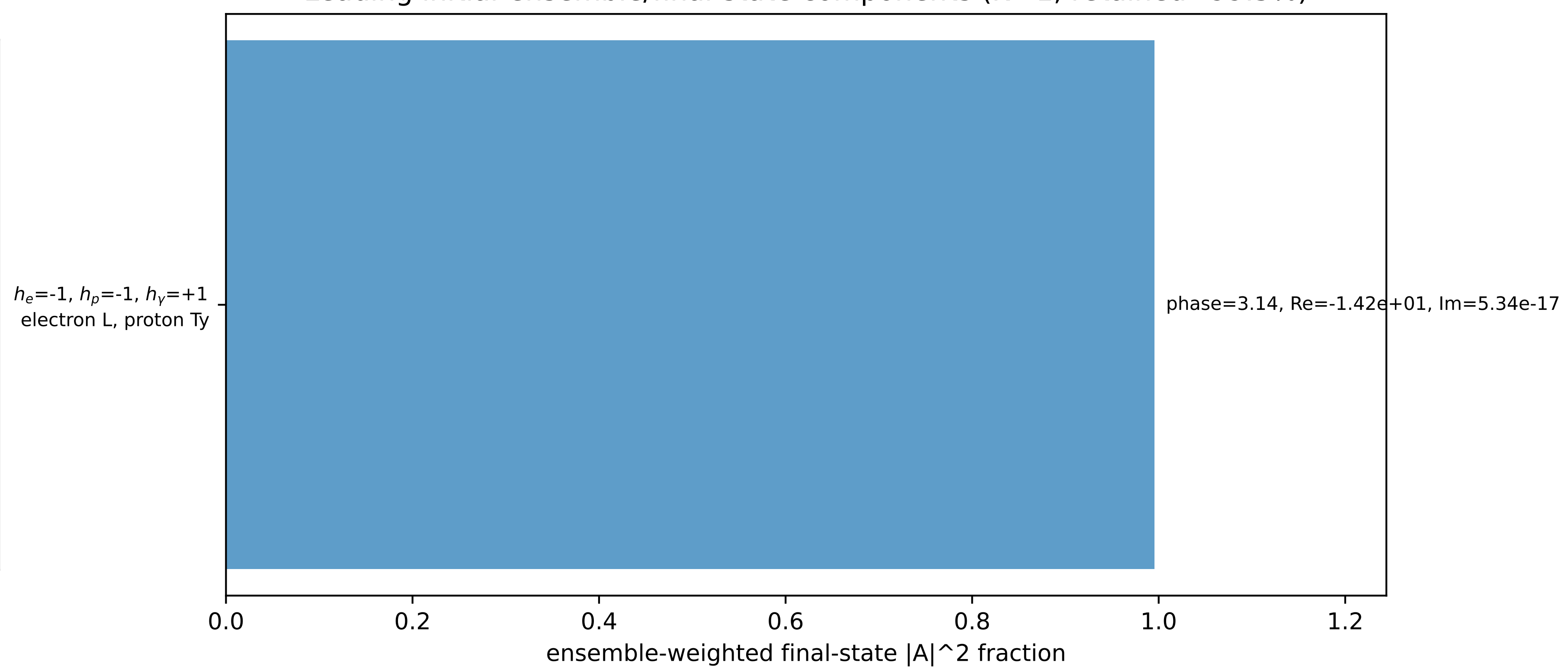
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=1.96296
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=1.5708
Q^2=19.6902, x_B=0.99463, t=-17.4772, W^2=0.986143, y=0.959318
$\theta(\ell', \gamma)$=180 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 0.0000 p_y= 2.2244 p_z= -0.0000
P E= 2.4141 p_x= -0.0000 p_y= -2.2244 p_z= 0.0000
ℓ' E= 2.2130 p_x= -0.0000 p_y= -2.2130 p_z= 0.0000
P' E= 2.1756 p_x= 0.0000 p_y= 1.9630 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.0000 p_y= 0.2500 p_z= 0.0000

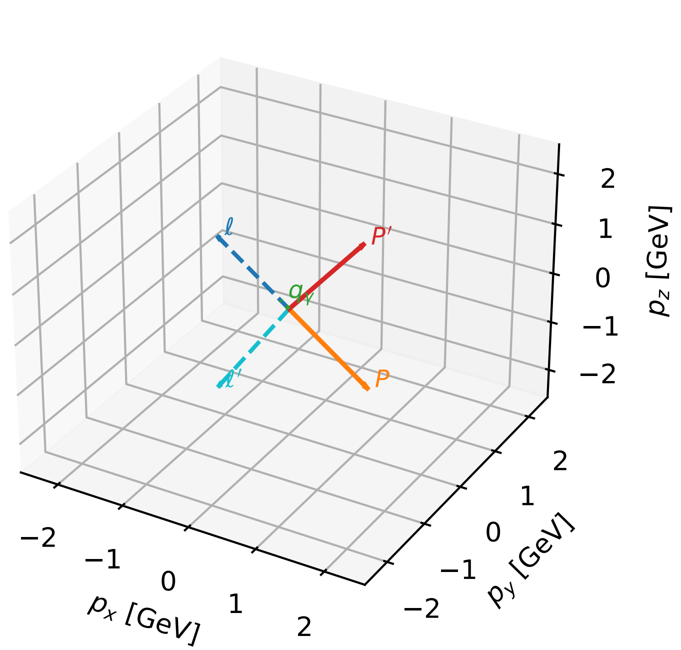
Transverse plane



Leading initial-ensemble/final-state components (N=1, retained=99.5%)



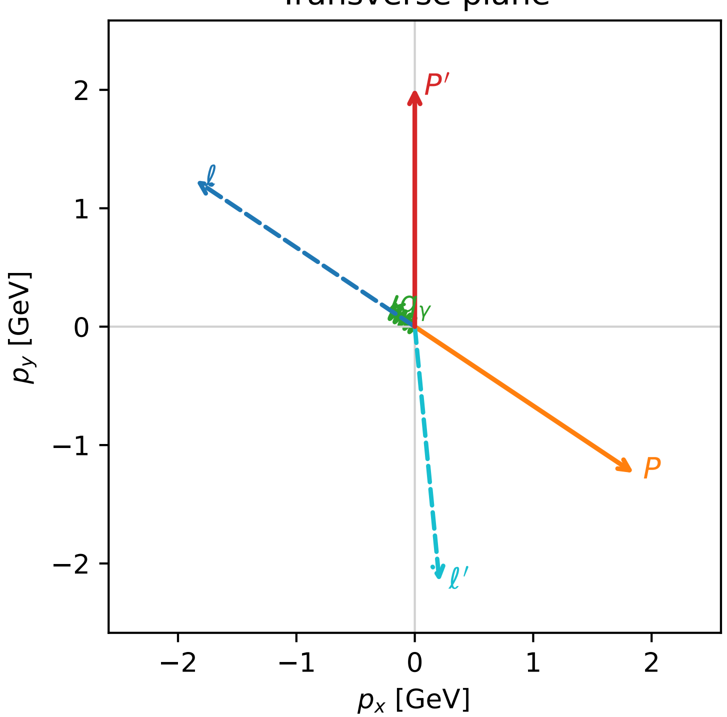
L electron + Ty proton characteristic max F_3 configuration

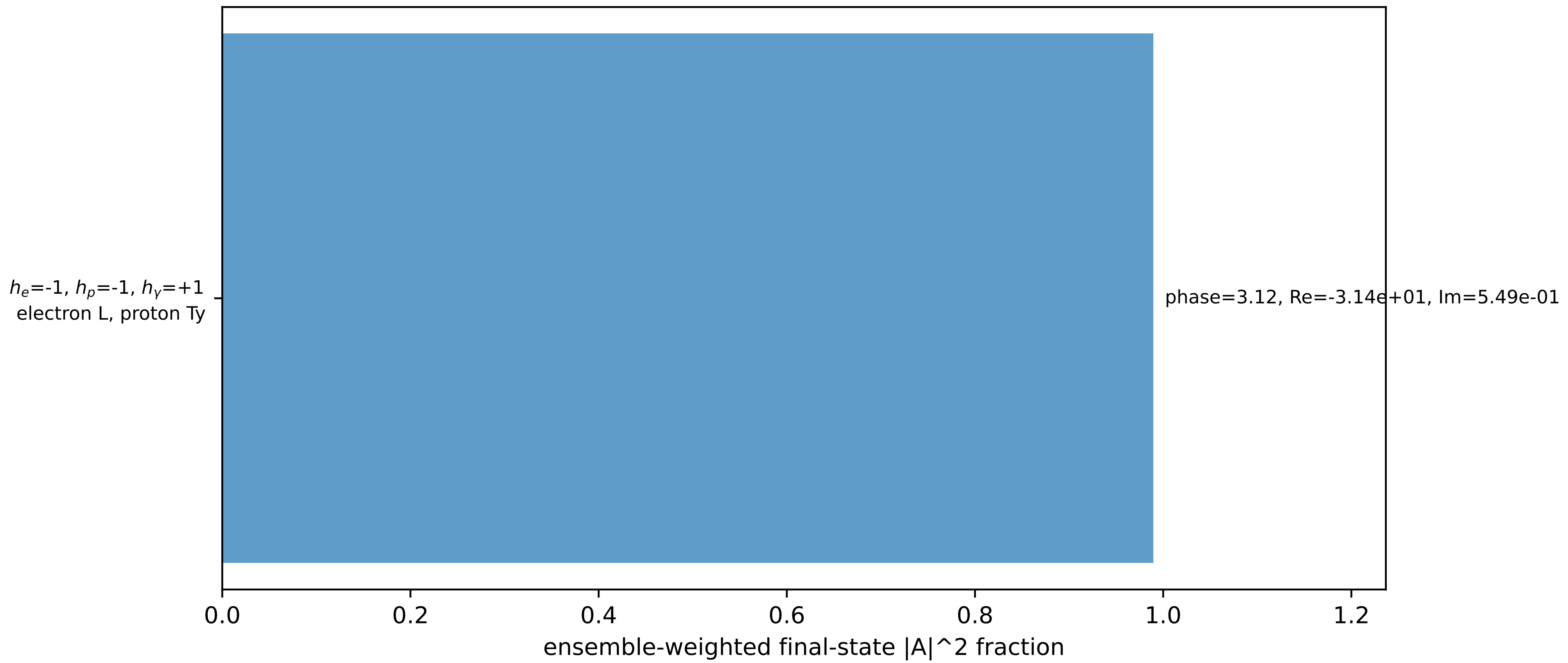



```
f3_low_egamma_lty_region_1  F_3=0.00910371
region: low_Egamma_LTy_region_1
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
```
incoming state: $|L_e\rangle \otimes |T_{y,p}\rangle$

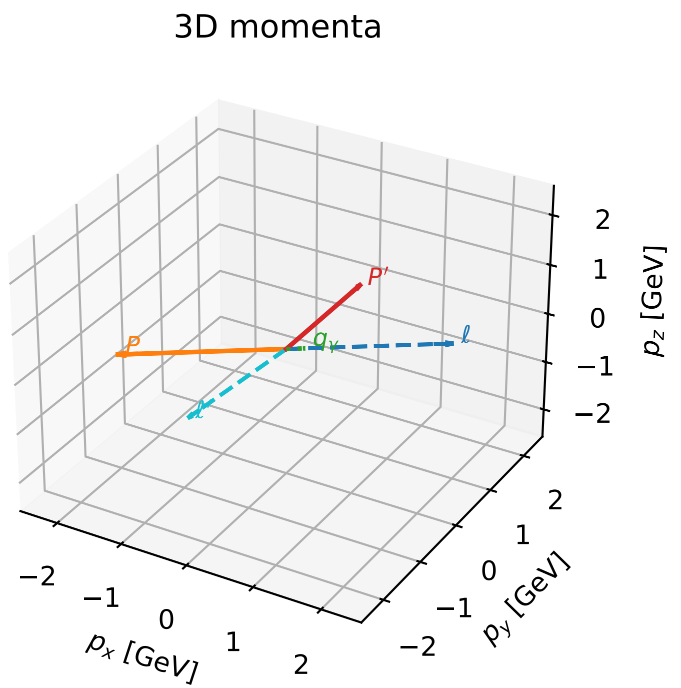
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.01605
$\theta_{\rm in}$=1.57079, ϕ_ℓ=2.55254, ϕ_P=5.69414
E_γ=0.25, ϕ_γ=2.55254
Q^2=15.7266, x_B=0.966105, t=-13.9591, W^2=1.4316, y=0.788834
$\theta(\ell', \gamma)$=129.26 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= 1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 2.1649 p_x= 0.2079 p_y= -2.1549 p_z= 0.0000
P' E= 2.2236 p_x= 0.0000 p_y= 2.0160 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2079 p_y= 0.1389 p_z= 0.0000





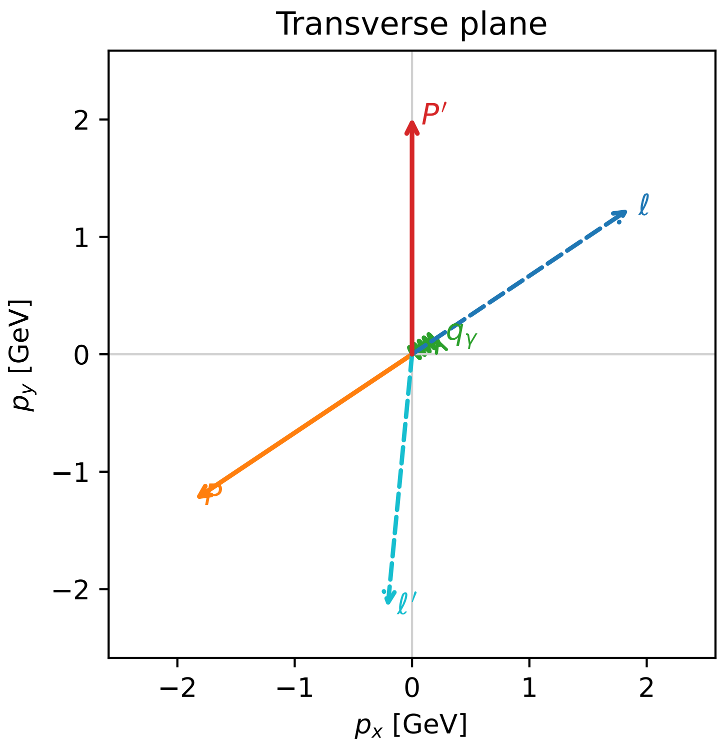
L electron + Ty proton characteristic max F_3 configuration

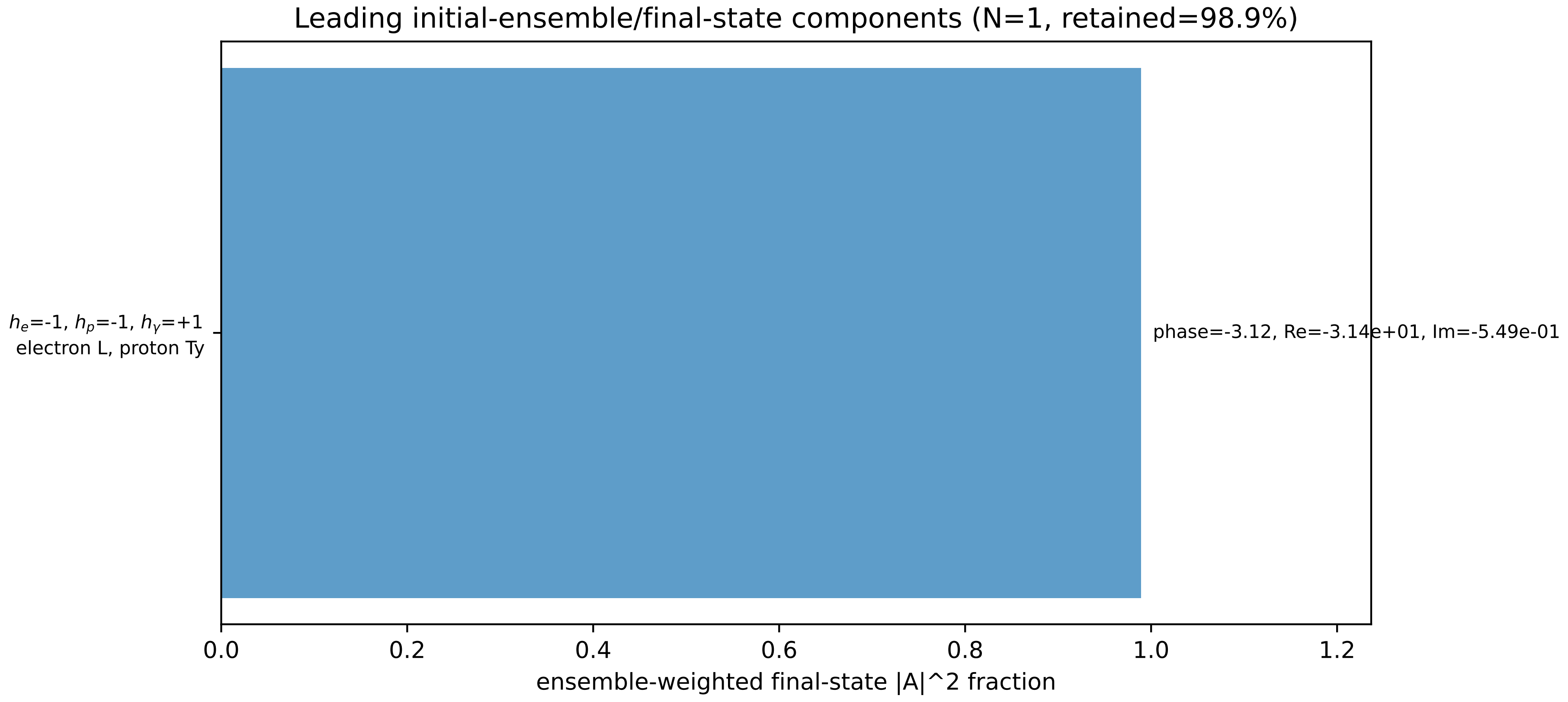


f3_low_egamma_lty_region_2 F_3=0.00910371
region: low_Egamma_LTy_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_e\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.01605
$\theta_{\rm in}$=1.57079, ϕ_ℓ=0.589049, ϕ_P=3.73064
E_γ=0.25, ϕ_γ=0.589049
Q^2=15.7266, x_B=0.966105, t=-13.9591, W^2=1.4316, y=0.788834
$\theta(\ell', \gamma)$=129.26 deg

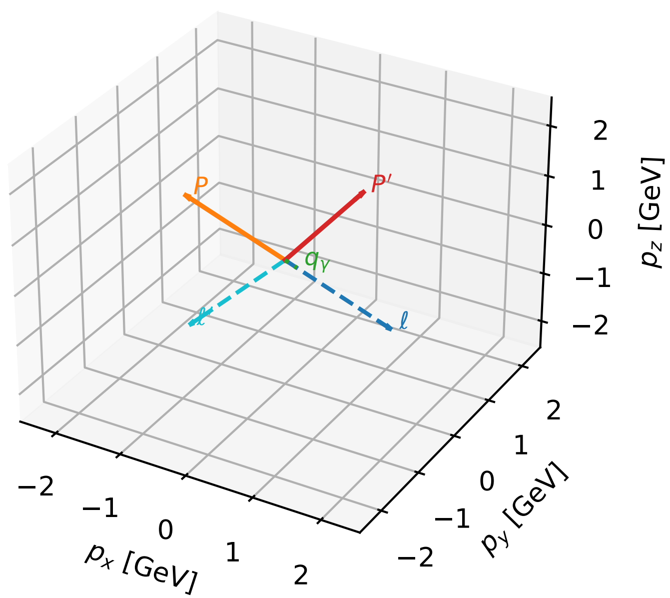
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 1.8495 p_y= 1.2358 p_z= -0.0000
P E= 2.4141 p_x= -1.8495 p_y= -1.2358 p_z= 0.0000
ℓ' E= 2.1649 p_x= -0.2079 p_y= -2.1549 p_z= 0.0000
P' E= 2.2236 p_x= 0.0000 p_y= 2.0160 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2079 p_y= 0.1389 p_z= 0.0000





L electron + Ty proton characteristic max F_3 configuration

3D momenta




```
f3_low_egamma_lty_region_3  F_3=0.00427465
region: low_Egamma_LTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
```

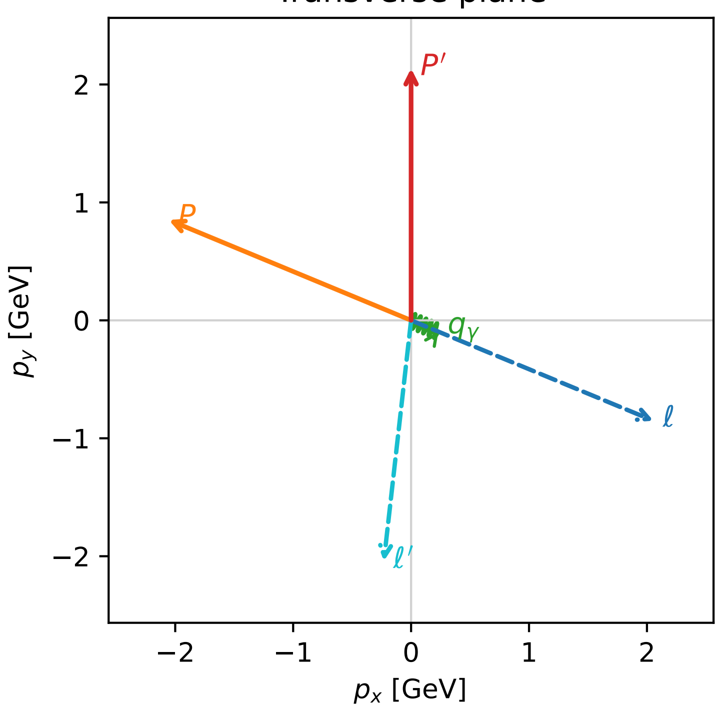
incoming state: $|L_e\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.13719
θ_{in}=1.57079, ϕ_ℓ=5.89049, ϕ_P=2.74889
E_γ=0.25, ϕ_γ=5.89049
Q^2=6.61398, x_B=0.807578, t=-5.87064, W^2=2.45576, y=0.396874
$\theta(\ell', \gamma)$=73.9548 deg

four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2244	p_x= 2.0551	p_y= -0.8512	p_z= -0.0000
P	E= 2.4141	p_x= -2.0551	p_y= 0.8512	p_z= 0.0000
ℓ'	E= 2.0545	p_x= -0.2310	p_y= -2.0415	p_z= 0.0000
P'	E= 2.3340	p_x= 0.0000	p_y= 2.1372	p_z= 0.0000
q_γ	E= 0.2500	p_x= 0.2310	p_y= -0.0957	p_z= 0.0000

Transverse plane



Leading initial-ensemble/final-state components (N=2, retained=99.7%)